Quiz 4: Linear maps

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Exercise 2. (4 Points) Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear map with

$$F\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}3\\5\\-2\end{pmatrix}, \qquad F\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\4\\-2\end{pmatrix}.$$

Calculate $F\begin{pmatrix}3\\1\end{pmatrix}$.

Solution.

$$F\begin{pmatrix}x_1\\x_2\end{pmatrix} = F\left(a\begin{pmatrix}2\\1\end{pmatrix} + b\begin{pmatrix}1\\1\end{pmatrix}\right)$$

$$\therefore \quad \begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{pmatrix}2 & 1\\1 & 1\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix}$$

Solving for $\begin{pmatrix}a\\b\end{pmatrix}$

$$\therefore \quad \left(\begin{array}{cc|c} 2 & 1 & x_1 \\ 1 & 1 & x_2 \end{array}\right)$$

$$\sim \quad \left(\begin{array}{cc|c} 1 & 0 & x_1 - x_2 \\ 1 & 1 & x_2 \end{array}\right)$$

$$\sim \quad \left(\begin{array}{cc|c} 1 & 0 & x_1 - x_2 \\ 0 & 1 & 2x_2 - x_1 \end{array}\right)$$

$$\therefore \quad a = x_1 - x_2$$
$$b = 2x_2 - x_1$$

$$F\begin{pmatrix}x_1\\x_2\end{pmatrix} = (x_1 - x_2)\,F\begin{pmatrix}2\\1\end{pmatrix} + (2x_2 - x_1)\,F\begin{pmatrix}1\\1\end{pmatrix}$$

$$\Rightarrow \quad F\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1 - x_2)\begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} + (2x_2 - x_1)\begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} 3x_1 - 3x_2 \\ 5x_1 - 5x_2 \\ -2x_1 + 2x_2 \end{pmatrix} + \begin{pmatrix} 2x_2 - x_1 \\ 8x_2 - 4x_1 \\ -4x_2 + 2x_1 \end{pmatrix}$$

$$= \begin{pmatrix} 2x_1 - x_2 \\ x_1 + 3x_2 \\ -2x_2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -1 \\ 1 & 3 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\therefore \quad F\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 - 1 \\ 3 + 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ -2 \end{pmatrix} \qquad - \ \underline{\underline{\text{Ans}}}$$